Package 'fieldborders'

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Type Package

Title Agricultural Field Boundaries from Sentinel-2 Time Series

Version 1.0.0

Description Delineates agricultural fields from Sentinel-2 image time series. The temporal boundary contrast method ranks spectral bands and indices by how differently they change through the season on the two sides of a field border and inside one field, learns an edge map from the selected contrasts and a cropland map from per-pixel temporal statistics, and turns both maps into field objects by marker-controlled watershed. The maps can be fused with the class probabilities of deep segmentation models such as the Fields of The World PRUE models, which can be run from R. Functions cover monthly Sentinel-2 download from a STAC catalogue, training with reference polygons, pretrained models, threshold tuning, polygon export, object-level accuracy assessment compatible with the Fields of The World benchmark, and maps of the results.

License GPL (>= 3)

URL https://github.com/ehsanrahimi666/fieldborders

BugReports https://github.com/ehsanrahimi666/fieldborders/issues

Depends R (>= 4.1.0)

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sf,
stats,
terra (>= 1.7-0),
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LinkingTo Rcpp

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`fb_accuracy` *Accuracy metrics from evaluation counts*

Description

Pools the counts of `fb_evaluate()` over tiles (optionally by group) and computes pixel precision, recall and IoU and object precision, recall and F1.

Usage

```
fb_accuracy(counts, by = NULL)
```

Arguments

`counts`	Data frame of counts, one row per tile, possibly with grouping columns.
`by`	Grouping column names, or `NULL` for one pooled summary.

Value

A data frame of metrics; `ref_s<k>` gives the number of reference fields in size class `k` and `recall_s<k>` their recall (`NA` if the class holds no field).

Examples

```
s <- fb_example_stack("california")
f <- fb_example_files("california")
ref <- fb_reference(sf::st_read(f$fields, quiet = TRUE), s)
# fields from the PRUE probabilities of the example, with the thresholds of the fusion model
dl <- fb_read_probabilities(f$prue)
th <- fb_model_pretrained("california", "fusion")$thresholds
tuned <- fb_delineate(dl, thresholds = th)
unfiltered <- fb_delineate(dl, thresholds = th, min_size = 0)
ev <- rbind(cbind(method = "tuned thresholds", fb_evaluate(ref, tuned)),
            cbind(method = "no size filter", fb_evaluate(ref, unfiltered)))
fb_accuracy(ev, by = "method")
```

fb_auc *Area under the ROC curve*

Description

Probability that a random positive scores higher than a random negative (ties count one half), computed from ranks.

Usage

```
fb_auc(score, y)
```

Arguments

score	Numeric scores.
y	Binary labels (1 positive, 0 negative).

Value

The AUC, or NA if one class is missing.

Examples

```
fb_auc(c(0.1, 0.4, 0.35, 0.8), c(0, 0, 1, 1))
```

fb_bootstrap *Paired bootstrap of accuracy differences between methods*

Description

Tiles are resampled with replacement (within strata, if given); the same resampled tiles are used for both methods of a comparison.

Usage

```
fb_bootstrap(
  counts,
  comparisons,
  metric = c("obj_f1", "obj_recall", "obj_precision"),
  B = 2000L,
  seed = 1L
)
```

Arguments

counts	Data frame with columns tile, method, optionally stratum, and the count columns of fb_evaluate().
comparisons	Two-column matrix (or a length-2 vector) of method names; the difference is first minus second.
metric	"obj_f1", "obj_recall" or "obj_precision".
B	Number of bootstrap resamples.
seed	Random seed.

Value

A data frame with the observed difference and the 2.5 and 97.5 percentiles of the bootstrap differences.

Examples

```
s <- fb_example_stack("california")
f <- fb_example_files("california")
ref <- fb_reference(sf::st_read(f$fields, quiet = TRUE), s)
m <- fb_model_pretrained("general")
fu <- fb_model_pretrained("general", "fusion")
maps <- fb_tbc_maps(s, m)
fused <- fb_fuse(maps, fb_read_probabilities(f$prue), fu)
fields <- list(TBC = fb_delineate(maps, thresholds = m$thresholds),
               FUSION = fb_delineate(fused, thresholds = fu$thresholds))
# the example window cut into four quarters, standing in for tiles
e <- as.vector(terra::ext(ref)); xm <- mean(e[1:2]); ym <- mean(e[3:4])
quarters <- list(c(e[1], xm, ym, e[4]), c(xm, e[2], ym, e[4]),
                 c(e[1], xm, e[3], ym), c(xm, e[2], e[3], ym))
counts <- do.call(rbind, lapply(seq_along(quarters), function(k) {
  q <- terra::ext(quarters[[k]])
  do.call(rbind, lapply(names(fields), function(nm)
    cbind(tile = k, method = nm, fb_evaluate(terra::crop(ref, q), terra::crop(fields[[nm]],
}))
fb_bootstrap(counts, c("FUSION", "TBC"), B = 500)
```

---

fb_components *Connected components*

Description

Labels connected groups of pixels that share the same positive value.

Usage

```
fb_components(x, connectivity = 4L)
```

Arguments

x	Logical or integer matrix; for integer input, pixels with different positive values are never joined.
connectivity	4 or 8.

Value

Integer matrix of component labels (0 = none).

Examples

```
m <- matrix(c(1, 1, 0, 0, 0, 1, 1, 0, 1), 3)
fb_components(m > 0)
fb_components(m > 0, connectivity = 8)
# cropland patches of the Washington example
maps <- fb_tbc_maps(fb_example_stack("washington"), fb_model_pretrained())
crop <- terra::as.matrix(maps[["cropland"]], wide = TRUE) > 0.5
max(fb_components(crop))
```

---

fb_dates *Acquisition dates of a stack*

Description

Acquisition dates of a stack

Usage

```
fb_dates(x)
```

Arguments

x	An fb_stack.

Value

A Date vector.

Examples

```
fb_dates(fb_example_stack("california"))
```

`fb_delineate` *Delineate field objects from edge and cropland maps*

Description

The object procedure validated with the package: markers are the 4-connected components of `{cropland > t_field and edge < t_edge}` with at least `min_marker` pixels; a marker-controlled watershed on the edge map, within `{cropland > 0.5}`, assigns the remaining cropland pixels; touching objects are separated by one-pixel lines and objects smaller than `min_size` pixels are removed.

Usage

```
fb_delineate(
  maps,
  t_edge = NULL,
  t_field = NULL,
  min_size = NULL,
  min_area = NULL,
  min_marker = 20L,
  thresholds = NULL
)
```

Arguments

`maps`	A `SpatRaster` with layers `edge` and `cropland` (`fb_tbc_maps()`, `fb_fuse()`) or deep-model probabilities with layers `background`, `field` and `boundary` (then `edge` = `P(boundary)` and `cropland` = `P(field) + P(boundary)`).
`t_edge, t_field`	Marker thresholds.
`min_size`	Minimum object size in pixels.
`min_area`	Optional minimum object area in square metres; overrides `min_size`.
`min_marker`	Minimum marker size in pixels.
`thresholds`	Optional list with `t_edge`, `t_field`, `min_size` and optionally `min_area` (for example the `thresholds` element of a model), used for arguments that are not given; a `min_area` in the list takes precedence over its `min_size`, so that the minimum field size does not depend on the pixel size.

Value

A `SpatRaster` (`field_id`) of field objects, 0 outside fields.

Examples

```
s <- fb_example_stack("california")
m <- fb_model_pretrained("california")
maps <- fb_tbc_maps(s, m)
fields <- fb_delineate(maps, thresholds = m$thresholds)
fb_plot_fields(fields, s, date = 7)
# thresholds can also be given one by one, and the minimum size as an area
fields2 <- fb_delineate(maps, t_edge = 0.3, t_field = 0.7, min_area = 5000)
```

---

fb_evaluate *Object-level accuracy of field delineation*

Description

Compares predicted field objects with reference fields using the object metric of the Fields of The World benchmark. Reference objects are the 4-connected components of field pixels; predicted objects are the 4-connected components of each label. A reference and a predicted object match when their intersection over union exceeds `iou` (with `iou >= 0.5` every match is one-to-one). Unmatched predicted objects with more than half of their pixels unlabelled (class 3) are not counted as false positives (`obj_fp`); `obj_fp_all` counts all of them.

Usage

```
fb_evaluate(
  reference,
  fields,
  iou = 0.5,
  pixel_area = NA_real_,
  size_breaks_ha = c(0, 1, 5, 20, 50, Inf),
  field = 1L,
  unlabelled = 3L
)
```

Arguments

`reference`	Reference classes: a raster from `fb_reference()` (its `class` layer is used) or an integer matrix (0 background, 1 field, 2 boundary, 3 unlabelled).
`fields`	Predicted objects: a raster from `fb_delineate()` or an integer matrix.
`iou`	Matching threshold (at least 0.5).
`pixel_area`	Pixel area in square metres for the size classes; computed from the raster when `reference` is a `SpatRaster`.
`size_breaks_ha`	Breaks of the reference-field size classes in hectares.
`field, unlabelled`	Class codes.

Value

A one-row data frame of counts: pixel `tp`, `fp`, `fn`; objects `obj_tp`, `obj_fp`, `obj_fp_all`, `obj_fn`, `n_pred`; and reference fields and matches per size class (`ref_s<k>`, `hit_s<k>`). Summarise with `fb_accuracy()`.

Examples

```
s <- fb_example_stack("california")
f <- fb_example_files("california")
ref <- fb_reference(sf::st_read(f$fields, quiet = TRUE), s)
m <- fb_model_pretrained("california")
fields <- fb_delineate(fb_tbc_maps(s, m), thresholds = m$thresholds)
ev <- fb_evaluate(ref, fields)
ev
fb_accuracy(ev)
```

---

fb_example_files *Example data shipped with the package*

Description

Small Sentinel-2 time series (for each month, the scene with the fewest invalid pixels in the tile that holds the window; ten bands, resampled to the Fields of The World grid of about 5 m, about 1 km across) of four agricultural landscapes, with the output of the PRUE deep model for the same window (`fb_prue_predict()` on the example stack), the Fields of The World global field map of the same year where available, and reference field polygons where their licence allows redistribution:

Usage

```
fb_example_files(name = fb_example_names())

fb_example_stack(name = fb_example_names())

fb_example_bands()

fb_example_names()
```

Arguments

`name`	Example name, one of `fb_example_names()`.

Details

- `"california"`: Colusa County, Sacramento Valley, USA, 2024: rice, orchards and row crops. Reference fields from the California Department of Water Resources Statewide Crop Mapping 2024 (CC0-1.0). PRUE months March and July.

- "washington": Columbia Basin, Washington, USA: centre-pivot and rectangular irrigated fields. PRUE months March and July.
- "brazil": Tocantins, Brazil, season July 2019 to June 2020: large summer-crop fields in the Cerrado. PRUE months September and January.
- "moghan": Moghan plain, Ardabil, Iran, 2024: narrow strip fields. PRUE months March and July.

The windows lie in tiles that were not used to fit or tune the pretrained models. Contains modified Copernicus Sentinel data 2019, 2020 and 2024. The FTW global field map is licensed CC-BY-4.0 (Fields of The World).

Value

fb_example_files(): a list with the image and clear-mask files, the dates, and the paths of the reference fields (fields), PRUE probabilities (prue) and FTW map (ftw); missing items are NULL. fb_example_stack(): an fb_stack. fb_example_bands(): the band names of the example images. fb_example_names(): the example names.

Examples

```
fb_example_names()
s <- fb_example_stack("washington")
s
fb_plot_rgb(s, date = 7)
str(fb_example_files("california"))
```

---

fb_feature_set *Candidate spectral features*

Description

Defines the per-date features whose temporal contrast across field borders is ranked by fb_rank_contrast(). The default set, used to develop and validate the method, holds the bands and the normalised differences of all band pairs, (a - b) / (a + b + 1e-4). With set = "svid", published spectral indices from the catalogue of the svid package (and, optionally, expressions generated by an svid grammar) that can be computed from the available bands are added.

Usage

```
fb_feature_set(bands, set = c("nd", "svid"), grammar = NULL)
```

Arguments

bands	Band names of the stack (Sentinel-2 names such as "B04").
set	"nd" (bands and normalised differences) or "svid".
grammar	Optional svid_grammar object whose expressions are added when set = "svid"; keep it small, because every expression is ranked.

Value

A data frame with columns `name`, `expr` (an R expression in the band names) and `type` (`"band"`, `"nd"`, `"svid"`).

Examples

```
fs <- fb_feature_set(c("B02", "B03", "B04", "B08"))
fs
if (requireNamespace("svid", quietly = TRUE)) {
  sv <- fb_feature_set(fb_example_bands(), set = "svid")
  table(sv$type)
  head(sv[sv$type == "svid", ])
}
```

---

fb_fields *Field boundaries in one step*

Description

Runs the complete workflow on a stack: TBC edge and cropland maps from a model, optional fusion with deep-model probabilities, field objects and polygons.

Usage

```
fb_fields(
  stack,
  model = fb_model_pretrained("general"),
  dl = NULL,
  fusion = if (is.null(dl)) NULL else fb_model_pretrained("general", "fusion"),
  thresholds = NULL,
  polygons = TRUE,
  threads = 1L
)
```

Arguments

`stack`	An `fb_stack`.
`model`	An `fb_model` (default: the pretrained `"general"` model).
`dl`	Optional deep-model probabilities (`background`, `field`, `boundary`), for example from `fb_prue_predict()`, on the grid of `stack`.
`fusion`	An `fb_fusion` model, used when `dl` is given (default: the pretrained `"general"` fusion).
`thresholds`	Delineation thresholds; by default those stored with the model (or the fusion, when `dl` is given).
`polygons`	Return polygons as well as rasters.
`threads`	Number of threads for the cropland model.

Value

A list with `maps` (edge and cropland), `fields` (field objects) and, if requested, `polygons` (`sf`).

Examples

```
s <- fb_example_stack("moghan")
res <- fb_fields(s)
res$polygons
fb_plot_fields(res$fields, s)
```

`fb_finalise` *Final field objects from a label matrix*

Description

Keeps labelled pixels flagged in `keep`, closes one-pixel cracks enclosed by a single label, and separates touching objects by a one-pixel line so that every object is a separate 4-connected component.

Usage

```
fb_finalise(labels, keep)
```

Arguments

`labels`	Integer label matrix.
`keep`	Logical matrix of pixels that belong to objects.

Value

Integer label matrix of final objects.

Examples

```
lab <- matrix(rep(1:2, each = 50), 10, 10)
table(fb_finalise(lab, lab > 0))
# watershed basins of the Washington example turned into separate objects
maps <- fb_tbc_maps(fb_example_stack("washington"), fb_model_pretrained())
E <- terra::as.matrix(maps[["edge"]], wide = TRUE)
F <- terra::as.matrix(maps[["cropland"]], wide = TRUE)
ws <- fb_watershed(E, fb_components(F > 0.7 & E < 0.3), mask = F > 0.5)
obj <- fb_finalise(ws, F > 0.5)
max(fb_components(obj))
```

`fb_fit_fusion` *Fit a fusion of TBC maps and deep-model probabilities*

Description

Two logistic regressions are fitted on reference pixels: boundary versus field interior on `logit P(boundary)` of the deep model and `logit(edge)` of TBC, and field or boundary versus background on `logit P(field + boundary)` and `logit(cropland)`, each with a ridge penalty of 1 on the slopes.

Usage

```
fb_fit_fusion(tbc_maps, dl_maps, references, n_pixels = 20000L, seed = 1L)
```

Arguments

`tbc_maps`	A `SpatRaster` from `fb_tbc_maps()`, or a list of them.
`dl_maps`	Deep-model probabilities with layers `background`, `field`, `boundary` (`fb_prue_predict()`, `fb_read_probabilities()`), or a list of them.
`references`	Reference rasters from `fb_reference()`, or a list of them.
`n_pixels`	Maximum number of pixels per class and tile.
`seed`	Random seed for the pixel sample.

Value

An object of class `fb_fusion`.

Examples

```
s <- fb_example_stack("california")
f <- fb_example_files("california")
ref <- fb_reference(sf::st_read(f$fields, quiet = TRUE), s)
tbc <- fb_tbc_maps(s, fb_model_pretrained("california"))
dl <- fb_read_probabilities(f$prue)
fu <- fb_fit_fusion(tbc, dl, ref, n_pixels = 5000)
fu
```

fb_fuse *Fuse TBC maps with deep-model probabilities*

Description

Fuse TBC maps with deep-model probabilities

Usage

```
fb_fuse(tbc_maps, dl_maps, fusion)
```

Arguments

tbc_maps	A SpatRaster from fb_tbc_maps().
dl_maps	Deep-model probabilities (background, field, boundary) on the same grid.
fusion	An fb_fusion object from fb_fit_fusion() or fb_model_pretrained().

Value

A SpatRaster with fused edge and cropland layers, ready for fb_delineate().

Examples

```
s <- fb_example_stack("california")
f <- fb_example_files("california")
tbc <- fb_tbc_maps(s, fb_model_pretrained("california"))
fused <- fb_fuse(tbc, fb_read_probabilities(f$prue), fb_model_pretrained("california", "fusi
fb_plot_maps(fused)
```

fb_model_pretrained *Pretrained models*

Description

Models trained with fb_train() and fb_fit_fusion() on the reference tiles of the validation, with delineation thresholds tuned on separate tiles of the same regions:

Usage

```
fb_model_pretrained(
  region = c("general", "california", "ftw"),
  type = c("tbc", "fusion")
)
```

Arguments

region	Model name.
type	"tbc" for the TBC model, "fusion" for the fusion with the PRUE B3 model (not available for "ftw").

Details

- "general" (default): fitted on 44 tiles in California (2024) and 31 tiles in Washington (2024 and 2025) together, thresholds tuned on 22 and 15 further tiles; the model to use where no local reference fields exist.
- "california": fitted on the 44 California tiles, thresholds tuned on 22.
- "ftw": fitted on 89 Fields of The World benchmark chips from six countries, using the four 10 m bands only, thresholds tuned on the California and Washington tiles; in the validation, a model fitted this way transferred to annual crops but not to perennial crops.

The cropland learner of the "general" and "california" models was chosen among ridge logistic regression, random forest and gradient-boosted trees by object F1 on the tuning tiles (gradient-boosted trees in both); the "ftw" model uses gradient-boosted trees. The models were fitted on Sentinel-2 images resampled to the Fields of The World chip grid (about 5 m); the minimum field size is stored in pixels and as an area (min_area, square metres), which fb_delineate() uses, so that it keeps its meaning at other pixel sizes. In all models the tuned minimum field size is about 1.1 ha, the largest value of the tuning grid, which suits the large fields of the tuning regions; where fields are smaller, pass a smaller min_area to fb_delineate(). The fusion models combine TBC with PRUE B3 probabilities from the March and July scenes. The test tiles of the validation were not used to fit or tune any model.

Value

An fb_model or fb_fusion object.

Examples

```
fb_model_pretrained()
fb_model_pretrained("california", "fusion")
```

---

fb_plot_accuracy *Accuracy of several methods*

Description

Accuracy of several methods

Usage

```
fb_plot_accuracy(
  acc,
  type = c("objects", "size"),
  size_labels = c("< 1", "1-5", "5-20", "20-50", "> 50")
)
```

Arguments

acc	Output of fb_accuracy() with a method column (one row per method).
type	"objects" for object precision, recall and F1, or "size" for the recall of reference fields by size class (classes without reference fields are left out; the axis labels give the number of reference fields, from the first row of acc).
size_labels	Labels of the size classes (for type = "size").

Value

A ggplot object.

Examples

```
s <- fb_example_stack("california")
f <- fb_example_files("california")
ref <- fb_reference(sf::st_read(f$fields, quiet = TRUE), s)
m <- fb_model_pretrained("california")
fu <- fb_model_pretrained("california", "fusion")
maps <- fb_tbc_maps(s, m)
dl <- fb_read_probabilities(f$prue)
fields <- list(TBC = fb_delineate(maps, thresholds = m$thresholds),
               `PRUE B3` = fb_delineate(dl, thresholds = fu$thresholds),
               `TBC + PRUE B3` = fb_delineate(fb_fuse(maps, dl, fu), thresholds = fu$thresho
ev <- do.call(rbind, lapply(names(fields), function(nm)
  cbind(method = nm, fb_evaluate(ref, fields[[nm]]))))
acc <- fb_accuracy(ev, by = "method")
fb_plot_accuracy(acc)
fb_plot_accuracy(acc, type = "size")
```

---

fb_plot_fields *Field objects or polygons over the image*

Description

Field objects or polygons over the image

Usage

```
fb_plot_fields(
  fields,
  stack = NULL,
  date = NULL,
  reference = NULL,
  colour = "#FFE14D",
  fill_alpha = 0.25,
  title = NULL
)
```

Arguments

fields	Field objects (SpatRaster from fb_delineate()) or polygons (sf).
stack	Optional fb_stack shown in true colour underneath.
date	Date of the background image (see fb_plot_rgb()).
reference	Optional reference polygons (sf), drawn as dashed outlines.
colour	Outline colour of the fields.
fill_alpha	Opacity of the field fill (0 for outlines only).
title	Plot title.

Value

A ggplot object.

Examples

```
s <- fb_example_stack("moghan")
m <- fb_model_pretrained()
fields <- fb_delineate(fb_tbc_maps(s, m), thresholds = m$thresholds)
fb_plot_fields(fields, s, date = 7)
# with reference fields (dashed)
s2 <- fb_example_stack("california")
f2 <- fb_delineate(fb_tbc_maps(s2, m), thresholds = m$thresholds)
ref_poly <- sf::st_read(fb_example_files("california")$fields, quiet = TRUE)
fb_plot_fields(f2, s2, date = 7, reference = ref_poly)
```

---

fb_plot_maps *Map layers (edge, cropland or class probabilities)*

Description

Map layers (edge, cropland or class probabilities)

Usage

```
fb_plot_maps(maps, layers = names(maps), palette = "magma", max_cells = 250000)
```

Arguments

maps	A SpatRaster with probability layers, such as the output of fb_tbc_maps(), fb_fuse() or fb_prue_predict().
layers	Layers to show (default all).
palette	Viridis palette option ("magma", "viridis", ...).
max_cells	Rasters with more cells are averaged in blocks for display.

Value

A `ggplot` object with one panel per layer.

Examples

```
maps <- fb_tbc_maps(fb_example_stack("california"), fb_model_pretrained("california"))
fb_plot_maps(maps)
fb_plot_maps(fb_read_probabilities(fb_example_files("brazil")$prue), palette = "viridis")
```

---

fb_plot_ranking *Feature ranking by border contrast*

Description

Feature ranking by border contrast

Usage

```
fb_plot_ranking(contrast, n = 20, selected = NULL)
```

Arguments

`contrast`	An `fb_contrast` object from `fb_rank_contrast()`.
`n`	Number of top features shown.
`selected`	Optional names of selected features, highlighted.

Value

A `ggplot` object.

Examples

```
s <- fb_example_stack("california")
f <- fb_example_files("california")
ref <- fb_reference(sf::st_read(f$fields, quiet = TRUE), s)
rk <- fb_rank_contrast(s, ref, n_pairs = 1000)
fb_plot_ranking(rk, n = 15, selected = fb_select_features(rk, k = 5))
```

`fb_plot_rgb` *True-colour image of one date*

Description

True-colour image of one date

Usage

```
fb_plot_rgb(
  stack,
  date = NULL,
  bands = c("B04", "B03", "B02"),
  stretch = c(0.02, 0.98),
  title = NULL,
  max_cells = 1e+06
)
```

Arguments

`stack`	An `fb_stack`.
`date`	Date (or index of the date) to show; by default the date with the most clear pixels.
`bands`	Red, green and blue bands.
`stretch`	Lower and upper quantiles of the linear stretch (common to the three bands).
`title`	Plot title; by default the date.
`max_cells`	Images with more cells are averaged in blocks for display.

Value

A `ggplot` object; `sf` layers can be added with `ggplot2::geom_sf()`.

Examples

```
fb_plot_rgb(fb_example_stack("california"), date = 7)
# false colour: near infrared, red, green
fb_plot_rgb(fb_example_stack("washington"), date = 7, bands = c("B08", "B04", "B03"))
```

fb_polygons *Field polygons from field objects*

Description

Converts field objects to polygons with their area, perimeter, compactness and, optionally, mean edge and cropland probability.

Usage

```
fb_polygons(fields, maps = NULL, simplify = 0, min_area = 0)
```

Arguments

fields	A SpatRaster of field objects from fb_delineate().
maps	Optional map SpatRaster whose layer means are added per field.
simplify	Simplification tolerance in map units (0 for none); in geographic coordinates a value around half a pixel removes the stair steps of pixel edges.
min_area	Minimum polygon area in square metres.

Value

An sf data frame with field_id, area_ha, perimeter_m, compactness (4 pi area / perimeter^2) and map means.

Examples

```
s <- fb_example_stack("california")
m <- fb_model_pretrained("california")
maps <- fb_tbc_maps(s, m)
p <- fb_polygons(fb_delineate(maps, thresholds = m$thresholds), maps)
head(p)
summary(p$area_ha)
```

fb_prue_download *Download the PRUE deep model for R*

Description

Downloads the Fields of The World PRUE EfficientNet-B3 model trained on CC-BY data (release v3.1), exported to TorchScript so that it runs with the R torch package without Python. The file (about 54 MB) is cached and verified by its MD5 checksum (SHA-256 9bd353b83826e84ab7c55173e9399674c778c729bf25

Usage

```
fb_prue_download(
  dir = tools::R_user_dir("fieldborders", "cache"),
  url = prue_url,
  overwrite = FALSE
)
```

Arguments

dir	Cache directory.
url	Download address of the TorchScript file (by default the release asset of the package on GitHub).
overwrite	Download again even if a verified copy exists.

Value

Path of the model file.

References

Kerner, H. et al. (2025) Fields of The World: a machine learning benchmark dataset for global agricultural field boundary segmentation. Proceedings of the AAAI Conference on Artificial Intelligence 39, 28151-28159. doi:10.1609/aaai.v39i27.35034

Examples

```
## Not run:
# downloads about 54 MB once; later calls return the cached file
f <- fb_prue_download()
f

## End(Not run)
```

---

fb_prue_predict *Run the PRUE deep model on a stack*

Description

Applies the Fields of The World PRUE model to two dates of a stack, as the model was trained: red, green, blue and near-infrared reflectance of the earlier date, then of the later date, divided by 0.3, rows ordered from south to north, 256-pixel windows with a stride of 128 pixels whose softmax probabilities are averaged. On the 54 California test tiles of the validation, its probabilities differed from those of the Python implementation by at most 1.2e-4 (mean absolute difference 7e-8).

Usage

```
fb_prue_predict(
  stack,
  months = c(3, 7),
  model = fb_prue_download(),
  window = 256L,
  stride = 128L,
  device = "cpu"
)
```

Arguments

`stack`	An `fb_stack` with bands `B04`, `B03`, `B02`, `B08`.
`months`	Calendar months of the two input dates, the earlier first in the season (for example `c(3, 7)` in the northern hemisphere, `c(9, 1)` in the southern). For each month the clearest date of that month is used, or of an adjacent month.
`model`	TorchScript model file from `fb_prue_download()`.
`window, stride`	Window size and stride in pixels.
`device`	`"cpu"` or `"cuda"`.

Details

The model sees each pixel in the context of its window, so probabilities depend on the surrounding image: areas smaller than a window are padded by reflection, and the output for a small area can differ from the output for the same pixels within a larger area. Prefer areas of at least 256 x 256 pixels (about 1.3 km at 5 m).

Value

A `SpatRaster` with layers `background`, `field` and `boundary`.

Examples

```
if (requireNamespace("torch", quietly = TRUE) && torch::torch_is_installed()) {
  s <- fb_example_stack("california")
  p <- fb_prue_predict(s, months = c(3, 7))     # downloads the model on first use
  fb_plot_maps(p)
}
```

`fb_rank_contrast` *Rank spectral features by their temporal contrast across field borders*

Description

For pixel pairs centred on reference field boundaries (across pairs: the two ends lie in different fields) and inside fields (within pairs), every candidate feature is summarised by the temporal maximum and mean, over dates clear at both ends, of `log(|f(p) - f(q)| + 1e-4)`. Features are ranked by the area under the ROC curve (AUC) with which the temporal maximum separates across from within pairs, averaged over strata. This extends the class-separability screening of spectral indices (as in the `svid` package) from classes to field borders.

Usage

```
fb_rank_contrast(
  stacks,
  references,
  features = NULL,
  strata = NULL,
  n_pairs = 4000L,
  seed = 1L
)
```

Arguments

`stacks`	An `fb_stack` or a list of them (training tiles). For many tiles, list elements may be functions without arguments that return the stack; each tile is then loaded only while it is processed.
`references`	A reference raster from `fb_reference()` or a list of them, one per stack (or functions returning them).
`features`	Candidate features from `fb_feature_set()`; by default the bands and all normalised differences of the first stack's bands.
`strata`	Optional stratum label per stack; the AUC is averaged over strata.
`n_pairs`	Maximum number of across and of within pairs per stack.
`seed`	Random seed for pair sampling.

Value

An object of class `fb_contrast` with `ranking` (data frame of features with their AUC, by stratum and averaged) and the pair summaries used by `fb_train()`.

Examples

```
s <- fb_example_stack("california")
f <- fb_example_files("california")
ref <- fb_reference(sf::st_read(f$fields, quiet = TRUE), s)
```

```
rk <- fb_rank_contrast(s, ref, n_pairs = 1000)
head(rk$ranking)
fb_plot_ranking(rk, n = 15)
```

`fb_read_probabilities` *Read deep-model class probabilities*

Description

Reads a three-layer raster of class probabilities (background, field, boundary), for example written by the Fields of The World tools or by `fb_prue_predict()`.

Usage

```
fb_read_probabilities(x, layers = c("background", "field", "boundary"))
```

Arguments

`x`	A raster file or `SpatRaster` with three layers.
`layers`	Order of the classes in `x`.

Value

A `SpatRaster` with layers `background`, `field` and `boundary`.

Examples

```
p <- fb_read_probabilities(fb_example_files("california")$prue)
p
```

`fb_read_stack` *Read an image time-series stack from files*

Description

Read an image time-series stack from files

Usage

```
fb_read_stack(files, bands, dates = NULL, clear = NULL, scale = 10000)
```

Arguments

files	One raster file per date, each holding bands in order.
bands	Band names in layer order.
dates	Acquisition dates; if NULL, they are read from the file names (YYYY-MM-DD or YYYYMMDD).
clear	Optional files with one clear-sky mask per date (1 clear, 0 not clear).
scale	Divisor that converts stored values to reflectance.

Value

An fb_stack.

Examples

```
f <- fb_example_files("california")
s <- fb_read_stack(f$images, bands = fb_example_bands(), clear = f$clear)
s
```

---

fb_reference *Reference raster from field polygons*

Description

Rasterizes reference field polygons onto the grid of a stack or raster. Field interiors are class 1, polygon outlines (burnt as one-pixel lines) class 2, and the remaining pixels background (class 0) or, when only some fields are mapped, unlabelled (class 3, ignored in training and in false-positive counts).

Usage

```
fb_reference(polygons, template, presence_only = FALSE)
```

Arguments

polygons	Reference fields as an sf object or a SpatVector of polygons, one feature per field.
template	An fb_stack or SpatRaster defining the grid.
presence_only	TRUE when pixels outside the polygons are not known to be background (for example when only one crop was mapped).

Value

A SpatRaster with layers class (0 background, 1 field, 2 boundary, 3 unlabelled) and field_id (polygon number, 0 outside fields).

Examples

```
f <- fb_example_files("california")
s <- fb_example_stack("california")
ref <- fb_reference(sf::st_read(f$fields, quiet = TRUE), s)
ref
terra::freq(ref$class)
```

`fb_s2_monthly` *Monthly Sentinel-2 time series for an area*

Description

Searches a STAC catalogue (by default Microsoft Planetary Computer) for Sentinel-2 Level-2A scenes, keeps per calendar month the scene with the smallest share of invalid pixels in the area (among the `max_per_month` scenes with the least scene-level cloud cover) (scene classification classes 0, 1, 3, 8, 9 and 10: no data, saturated, cloud shadow, medium and high cloud probability, thin cirrus), skips months with more than `max_invalid` invalid pixels, and resamples the bands bilinearly (the scene classification by nearest neighbour) to a common grid. The radiometric offset of processing baseline 04.00 and later is removed.

Usage

```
fb_s2_monthly(
  aoi,
  start,
  end,
  bands = c("B02", "B03", "B04", "B05", "B06", "B07", "B08", "B8A", "B11", "B12"),
  resolution = 5,
  max_cloud = 60,
  max_invalid = 0.2,
  max_per_month = 4L,
  invalid_classes = c(0, 1, 3, 8, 9, 10),
  stac_url = "https://planetarycomputer.microsoft.com/api/stac/v1",
  collection = "sentinel-2-l2a",
  dir = NULL,
  verbose = TRUE
)
```

Arguments

`aoi`	Area of interest: an `sf` or `SpatVector` object, a numeric bounding box `c(xmin, ymin, xmax, ymax)` in longitude and latitude, or a `SpatRaster` whose grid is used as is.
`start, end`	First and last date of the period (`"YYYY-MM-DD"`).
`bands`	Sentinel-2 bands to read.

`resolution` Pixel size in metres of the output grid (ignored when `aoi` is a `SpatRaster`); the grid uses the UTM zone of the area's centre. The pretrained models were developed on Sentinel-2 resampled to about 5 m (the Fields of The World chip grid), which is the default.

`max_cloud` Upper limit of the scene-level cloud cover (percent) in the search.

`max_invalid` Largest share of invalid pixels in the area for a month to be kept.

`max_per_month` Number of scenes per month, those with the least scene-level cloud cover, whose scene classification is read to find the clearest one.

`invalid_classes` Scene classification classes treated as invalid; add 11 to treat snow as invalid.

`stac_url`, `collection` STAC API and collection.

`dir` Optional directory where the stack is written (`fb_write_stack()`).

`verbose` Print progress.

Details

The scene classification misses some clouds, haze and cloud shadows, which then enter the series as clear pixels; look at the images (for example with `fb_plot_rgb()` for every date) and drop affected dates with `fb_subset_dates()`. In the southern hemisphere, choose a period that holds the summer growing season, for example July to June.

Value

An `fb_stack` of digital numbers (`scale` = `10000`) with clear-sky masks; the scene identifiers are stored in the attribute `"items"`.

Examples

```
## Not run:
# about 1.6 km x 1.6 km of centre-pivot fields near Jovein, Iran; one scene per month of
# 2024 (needs an internet connection; takes a few minutes)
s <- fb_s2_monthly(c(57.436, 36.606, 57.454, 36.620), "2024-01-01", "2024-12-31")
s
fb_plot_rgb(s, date = 7)

## End(Not run)
```

`fb_save_model` *Save and load a model*

Description

Save and load a model

Usage

```
fb_save_model(model, file)

fb_load_model(file)
```

Arguments

model	An fb_model or fb_fusion object.
file	Path of the .rds file.

Value

fb_save_model() returns the file path invisibly; fb_load_model() the model.

Examples

```
m <- fb_model_pretrained("california")
f <- tempfile(fileext = ".rds")
fb_save_model(m, f)
fb_load_model(f)
```

---

fb_select_features *Select features from a contrast ranking*

Description

Walks down the ranking and keeps a feature unless the absolute Spearman correlation of its temporal maximum with an already selected feature exceeds max_cor.

Usage

```
fb_select_features(contrast, k = 8L, max_cor = 0.9, n_sub = 100000L, seed = 1L)
```

Arguments

contrast	An fb_contrast object from fb_rank_contrast().
k	Number of features to select.
max_cor	Maximum absolute Spearman correlation between selected features.
n_sub	Number of pairs used for the correlations.
seed	Random seed for the subsample.

Value

Character vector of selected feature names, in rank order.

Examples

```
s <- fb_example_stack("california")
f <- fb_example_files("california")
ref <- fb_reference(sf::st_read(f$fields, quiet = TRUE), s)
rk <- fb_rank_contrast(s, ref, n_pairs = 1000)
fb_select_features(rk, k = 5)
```

`fb_stack` *Image time-series stack*

Description

Bundles a multi-date, multi-band image with its acquisition dates, band names and per-date clear-sky masks. All other functions of the package take this object.

Usage

```
fb_stack(x, dates, bands, clear = NULL, scale = 10000)
```

Arguments

`x`	A `SpatRaster` whose layers are the bands of each date, date by date (all bands of the first date, then all bands of the second date, and so on), or a list of `SpatRaster`s with one multi-band raster per date.
`dates`	Acquisition dates, one per date (`Date` or `"YYYY-MM-DD"` strings).
`bands`	Band names in layer order within each date, for example `c("B02", "B03", "B04", "B08")`. Sentinel-2 band names are expected by the pretrained models and the default feature set.
`clear`	Optional `SpatRaster` with one layer per date: 1 for clear pixels, 0 for cloud, cloud shadow, snow or no data. `NULL` treats all pixels as clear; pixels with missing band values are always treated as not clear.
`scale`	Divisor that converts stored values to surface reflectance (10000 for Sentinel-2 L2A digital numbers with the offset removed, 1 for reflectance).

Value

An object of class `fb_stack`: a list with `data` (`SpatRaster`), `clear` (`SpatRaster` or `NULL`), `dates`, `bands` and `scale`.

See Also

`fb_read_stack()`, `fb_s2_monthly()`, `fb_example_stack()`

Examples

```
# two dates of the California example, ten bands each, read with terra
f <- fb_example_files("california")
r <- terra::rast(f$images[c(3, 7)])
s <- fb_stack(r, dates = f$dates[c(3, 7)], bands = fb_example_bands(),
              clear = terra::rast(f$clear[c(3, 7)]))
s
fb_dates(s)
```

---

fb_subset_dates *Subset a stack by date*

Description

Subset a stack by date

Usage

```
fb_subset_dates(x, dates)
```

Arguments

x	An fb_stack.
dates	Dates to keep (Date or strings), or a logical or integer index.

Value

An fb_stack with the selected dates.

Examples

```
s <- fb_example_stack("california")
fb_subset_dates(s, format(s$dates) >= "2024-04-01" & format(s$dates) < "2024-10-01")
```

---

fb_tbc_maps *Edge and cropland maps from a TBC model*

Description

Computes, for every pixel, the edge probability (the maximum over four directions of the edge-model probability of the pixel pair centred on it) and the cropland probability. Large rasters are processed in blocks of rows.

Usage

```
fb_tbc_maps(stack, model, filename = "", block_cells = 100000L, threads = 1L)
```

Arguments

stack	An fb_stack holding the bands of the model.
model	An fb_model from fb_train() or fb_model_pretrained().
filename	Optional output GeoTIFF.
block_cells	Approximate number of cells per processing block.
threads	Number of threads for the cropland model.

Value

A SpatRaster with layers edge and cropland.

Examples

```
s <- fb_example_stack("california")
maps <- fb_tbc_maps(s, fb_model_pretrained("california"))
fb_plot_maps(maps)
```

---

fb_theme *Plot theme and colours of the package*

Description

Plot theme and colours of the package

Usage

```
fb_theme(base_size = 10)
```

Arguments

base_size	Base font size.

Value

A ggplot2 theme.

Examples

```
library(ggplot2)
ggplot(data.frame(x = 1:3, y = 1:3), aes(x, y)) + geom_point() + fb_theme()
```

`fb_train` *Train a temporal boundary contrast (TBC) model*

Description

Fits the two components of TBC on training tiles with reference fields:

- the edge model, a logistic regression of across versus within pairs on the temporal maximum and mean contrasts of the `k` selected features (standardised; missing summaries set to the training mean);
- the cropland model, gradient-boosted trees (`xgboost`), a probability random forest (`ranger`) or a logistic regression of field-interior versus background pixels on the 10th, 50th and 90th percentiles and the standard deviation over clear dates of the bands, NDVI, ND(B08,B11) and ND(B11,B12).

Usage

```
fb_train(
  stacks,
  references,
  features = NULL,
  strata = NULL,
  contrast = NULL,
  k = 8L,
  max_cor = 0.9,
  n_pairs = 4000L,
  n_pixels = 20000L,
  learner = c("xgboost", "ranger", "glm"),
  num_trees = 100L,
  min_node_size = 10L,
  thresholds = list(t_edge = 0.5, t_field = 0.5, min_size = 50L),
  seed = 1L,
  threads = 1L
)
```

Arguments

`stacks`	An `fb_stack` or a list of them (training tiles). For many tiles, list elements may be functions without arguments that return the stack; each tile is then loaded only while it is processed.
`references`	A reference raster from `fb_reference()` or a list of them, one per stack (or functions returning them).
`features`	Candidate features from `fb_feature_set()`; by default the bands and all normalised differences of the first stack's bands.
`strata`	Optional stratum label per stack; the AUC is averaged over strata.

contrast	Optional result of fb_rank_contrast() on the same data (avoids recomputing the pair summaries).
k, max_cor	Feature selection settings passed to fb_select_features().
n_pairs	Maximum number of across and of within pairs per stack.
n_pixels	Maximum number of field and of background pixels per stack for the cropland model.
learner	Cropland learner: "xgboost" (gradient-boosted trees, default, as in the validation of the method), "ranger" (random forest) or "glm" (ridge logistic regression).
num_trees	Number of trees (random forest) or boosting rounds (xgboost).
min_node_size	Minimum node size of the random forest; larger values give smaller models.
thresholds	Default object thresholds stored with the model (see fb_delineate()); tune them with fb_tune().
seed	Random seed for pair sampling.
threads	Number of threads for the cropland learner.

Details

The logistic regressions carry a ridge penalty of 1 on the slopes (the default of scikit-learn's LogisticRegression), which keeps them finite when the classes are separable.

Value

An object of class fb_model.

Examples

```
s <- fb_example_stack("california")
f <- fb_example_files("california")
ref <- fb_reference(sf::st_read(f$fields, quiet = TRUE), s)
m <- fb_train(s, ref, n_pairs = 1000, n_pixels = 2000, num_trees = 50)
m
```

---

fb_tune *Tune delineation thresholds on reference tiles*

Description

Evaluates every combination of thresholds on tiles with reference rasters and returns the object-level accuracy of each combination, pooled over tiles, with the best combination first.

Usage

```
fb_tune(
  maps,
  references,
  t_edge = c(0.2, 0.3, 0.4, 0.5, 0.6),
  t_field = c(0.5, 0.7, 0.9),
  min_size = c(0, 50, 100, 200),
  iou = 0.5
)
```

Arguments

`maps`	A map `SpatRaster` (see `fb_delineate()`) or a list of them.
`references`	Reference rasters from `fb_reference()`, one per map.
`t_edge`, `t_field`, `min_size`	Candidate values.
`iou`	Intersection-over-union required for a match.

Value

A data frame with one row per combination: thresholds, object precision, recall and F1, sorted by decreasing F1.

Examples

```
s <- fb_example_stack("california")
f <- fb_example_files("california")
ref <- fb_reference(sf::st_read(f$fields, quiet = TRUE), s)
maps <- fb_tbc_maps(s, fb_model_pretrained("california"))
tn <- fb_tune(maps, ref, t_edge = c(0.3, 0.5), t_field = c(0.5, 0.7), min_size = c(0, 50))
head(tn)
```

`fb_watershed` *Marker-controlled watershed*

Description

Floods `elev` from the markers in order of increasing value (4-connectivity), within `mask`.

Usage

```
fb_watershed(elev, markers, mask = NULL, lines = FALSE)
```

Arguments

elev	Numeric matrix (low values are field interiors, high values boundaries).
markers	Integer matrix of marker labels (0 = none).
mask	Logical matrix of pixels that may be labelled; defaults to all.
lines	If TRUE, pixels where basins meet are left unlabelled.

Value

Integer label matrix (0 = unlabelled).

Examples

```
# two basins separated by a ridge in column 10
e <- outer(1:20, 1:20, function(i, j) as.numeric(j == 10))
mk <- matrix(0L, 20, 20); mk[10, 5] <- 1L; mk[10, 15] <- 2L
table(fb_watershed(e, mk))
# fields of the Washington example: seeds inside fields, flooding of the edge map
maps <- fb_tbc_maps(fb_example_stack("washington"), fb_model_pretrained())
E <- terra::as.matrix(maps[["edge"]], wide = TRUE)
F <- terra::as.matrix(maps[["cropland"]], wide = TRUE)
seeds <- fb_components(F > 0.7 & E < 0.3)
ws <- fb_watershed(E, seeds, mask = F > 0.5)
length(unique(ws[ws > 0]))
```

---

fb_write_stack *Write an image time-series stack to files*

Description

Writes one GeoTIFF per date (bands in order, unsigned 16-bit when the stack holds digital numbers), one clear-mask GeoTIFF per date, and an index table.

Usage

```
fb_write_stack(x, dir, prefix = "s2")
```

Arguments

x	An fb_stack.
dir	Output directory (created if needed).
prefix	File name prefix.

Value

Invisibly, a data frame with the date and the image and mask paths.

Examples

```
s <- fb_example_stack("california")
d <- file.path(tempdir(), "stack")
idx <- fb_write_stack(s, d)
fb_read_stack(idx$image, bands = s$bands, dates = idx$date, clear = idx$clear)
```

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